

Occurrence, fate, and remediation of per- and polyfluoroalkyl substances in soils: A review

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Abstract

Per- and polyfluoroalkyl substances (PFAS) are a class of persistent organic contaminants and are found ubiquitously in the soil environment. These chemicals could enter agricultural food produce, and dietary consumption of these products potentially damages human health. This review summarizes the status of current knowledge regarding the sources of PFAS in soils, sorption and transport behaviors, and several remediation strategies. Land application of biosolids and utilization of aqueous firefighting foam formulations are the two major sources for PFAS to contaminate soils. PFAS sorption by soils undergoes multiple interaction mechanisms, which are related to PFAS chemical structures, soil properties, and the surrounding solution chemistry. The sorbed PFAS could migrate downward to groundwater. Although several remediation techniques have been tested at PFAS-contaminated sites, not much detailed information was documented; this limits the comparison and evaluation of the performance of the specific remediation technologies. Based on literature analysis, the current knowledge gaps are identified, and future research prospects suggested.

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Introduction

Per- and polyfluoroalkyl substances (PFAS) have been widely utilized in our household daily lives, industrial manufacturing, and aqueous firefighting foam (AFFF) formulations. Their extreme stability and high mobility lead to their ubiquitous distributions in the environment

and in humans. It is estimated that virtually 98% of Americans could have the detectable levels of PFAS in the blood [1]. Some regions and countries have adopted the maximum contaminant levels for perfluorooctanoic acid (PFOA), perfluorooctanesulfonic acid (PFOS) and other selected PFAS. These regulatory actions to safeguard human health currently focus primarily on drinking water exposure pathway. PFAS, considered a class of emerging contaminants, can be introduced to soils directly by applying AFFF or indirectly by reuse of treated wastewater in irrigation and land application of biosolids. PFAS in soils could be released to the surrounding water bodies, posing high risks to the ambient ecosystem health. These contaminants could enter agricultural produce (vegetables, food crops, and livestock meat), and dietary consumption of these produce represents an important exposure to human health. The corresponding feasible management protocols could be developed based on the new knowledge to mitigate PFAS accumulation in agricultural produce and to maintain sustainable agricultural production.

PFAS contamination in soils and water has been a major environmental threat to human health and agro-ecosystem services because of their environmental ubiquity, persistence, high mobility, bioaccumulation, and toxicity to humans and wildlife. PFAS are a family of anthropogenic chemicals containing a hydrophilic anionic functional group (e.g., carboxylate, or sulfonate) and a hydrophobic $-\text{CF}_2$ aliphatic chain. The hydrophilic nature leads to the relatively high water solubility hence mobility with water flow in soils. Since the discovery of PFAS in the 1930s, more than 13,000 PFAS chemicals have been utilized to manufacture carpets; clothing; furniture fabrics; food packaging; and water-, grease-, or stain-resistant cookware [2,3]. PFAS are also added in AFFF formulations and adopted in numerous industrial and commercial applications [4–8]. Owing to the strong covalent C–F bond, PFAS have extremely high thermal and chemical stability and are considered “forever chemicals.” While the head functional groups could be subject to transformations, the C–F bond, recognized as the strongest covalent bond in organic chemistry, is far more difficult to cleavage. The extreme persistence and high mobility of PFAS result in their ubiquitous distributions in the environment, wildlife, and humans [9–16]. PFAS with a long $-\text{CF}_2$ chain tend to be more bioaccumulative [12,17–20]. Many studies

showed that PFAS can interact with blood proteins and cause a number of pathological responses including hepatotoxicity, tumor induction, developmental toxicity, immunotoxicity, neurotoxicity, and endocrine disruption [21]. It was estimated that American adults on average take in up to 130 ng of PFOA per day from various exposure pathways [22], which could cause negative effects on human health, such as increasing serum lipids and liver enzymes, decreasing vaccine response, and reducing birth weight of newborns [23].

There have been an increasing number of studies to investigate sorption, transport, and remediation of PFAS in soils. Several recent comprehensive reviews have been published with focus on PFAS presence in biosolids and composted biosolids, bioavailability in soils, and remediation technologies [24,25]. A critical review is still necessary to update the status of the knowledge about PFAS occurrence in soils, sorption, transport, and conservative remediation technologies, and to guide future research efforts in these areas. The objective is to summarize the current research progress on sources, fate, sorption, transport, and remediation of PFAS in soils. Based on the reviewed literature, we highlight several knowledge gaps in these areas and recommend the corresponding perspectives for future research.

PFAS sources in soils

Soils are the major environmental reservoir for storing PFAS. There are multiple pathways for PFAS to enter soils, such as irrigation with PFAS-contaminated water, land application of biosolids, leachates from municipal waste landfills, use of AFFF formulations to extinguish fuel-based fires, and atmosphere deposition from industrial emissions [13,26–33]. The use of AFFF in either regular firefighting training or accidental fire suppression activities can cause PFAS discharge into soils and migration to the surrounding areas, leading to contamination of soil and subsurface environments as well as potential damages to the associated ecosystem health [4,34–37]. The applications of AFFF at firefighting training sites release PFAS generally within a limited area but at a relatively high concentration. For example, Brusseau et al. examined PFAS concentration in the soils from more than 1000 sites and found that PFOS was the predominant PFAS species at almost all sites. Perfluorohexanesulfonic acid (PFHxS), PFOA, and perfluorohexanoic acid (PFHxA) were among other PFAS frequently detected at the firefighting training sites [38]. The maximum concentration of PFOS in soils could reach 460,000 µg/kg, with the median concentration of 8722 µg/kg. The maximum reported PFOA concentration was 50,000 µg/kg, with the median value of 83 µg/kg [38]. Many other PFAS and their precursors were also found in the soils receiving the application of AFFF formulations. At a historical firefighting training site in Australia, more than 60 PFAS were found in the

AFFF-impacted soils, including perfluoroalkyl carboxylic acids (PFCA), perfluoroalkyl sulfonic acids (PFSA), and novel PFAS, such as chlorinated PFSA, ketone PFSA, perfluoroalkane sulfonamides, perfluoroalkyl sulfinic acids, perfluoroalkyl sulfonamidoethanols, fluorotelomer sulfonic acids, and perfluoroalkyl sulfonamide amines, as well as a fluorotelomer thioamido sulfonate [39].

Another major pathway for PFAS accumulation in soils is land application of biosolids and composted biosolids [24]. Biosolids are prepared from the sewage sludge produced in wastewater treatment processes and inherently contain high levels of PFAS and their precursors. Land application of biosolids is the common practice as a convenient waste management and resource recovery process to improve soil quality and fertility. Meanwhile, this practice disseminates a wide range of PFAS into agricultural production fields [26,40,41]. A national survey in 2007 estimated that about 7.2 million tons of biosolids (dry weight) were produced in the United States, and approximately 55% of the biosolids were land applied to soils [42]. PFAS concentration in biosolids could reach µg/kg and mg/kg levels [43–45]. In the United States, it was estimated that the total loadings of PFAS were 1.4–2.1 metric tons per year to agricultural fields solely from the land application of biosolids [46]. The repeated land application of biosolids could escalate PFAS concentration in soils. For example, the farmlands at Decatur, Alabama had received biosolids for more than one decade, as a result, PFAS concentrations in the soils were up to several mg/kg on the dry weight of soil [47]. The increasing accumulation of PFAS in soils is apparently correlated to the respective biosolids loading rates at the specific applied sites, independent on the patterns of historical biosolids applications [48,49]. Multiple PFAS are found to be distributed through the soil profiles, and short-chain PFAS demonstrate apparent leaching down in deep soils. Both PFOS and PFOA could be detected in the soils of the vadose zone at the depth of 17 m immediately adjunct to groundwater [48]. Many long-chain PFAS and the precursors are primarily retained in the solid fraction of biosolids. Degradation of the precursors during the sludge treatment processes and after land application to soils can lead to the increased concentration of short-chain PFAS [50]. PFAS originating from land application of biosolids have been reported to be taken up by food crops, which poses the risks to livestock and human health via dietary consumption of contaminated food crops [26,51–54].

Agricultural irrigation with PFAS-contaminated water is another source to disseminate these contaminants into soils [55]. The irrigation water generally is collected from groundwater (e.g., from wells) and/or surface water (e.g., from nearby streams, rivers, lakes, and ponds). This agricultural practice becomes increasingly common in the regions with irregular rainfall or drought periods during

growth season. This could be the consequence of global climate change with increasing duration and frequency of drought periods in many regions of the world. As a partial solution to the problem of water stress/shortage, the slightly contaminated freshwater, for example, secondary or tertiary treated wastewater, is being increasingly utilized to irrigate agricultural crops. PFAS are commonly found in the treated water at ng/L to µg/L levels [11,56]. The irrigation with treated wastewater can cause PFAS accumulation in soils with concentration of µg/kg levels [57,58]. Recently, Mroczko et al. analyzed 20 target PFAS in treated wastewater, well groundwater, and crops which were irrigated with the well water [59]. The identified PFAS were primarily PFCA and PFSA, and no target PFAS precursors were detected in water or crops. Ten PFAS chemicals were found with the total PFAS concentration of 88 ng L⁻¹ in the treated wastewater effluents, and the highest concentration was 155 ng L⁻¹ in the water from monitoring wells. PFAS concentration in well water showed little variation during different seasons. The highest PFAS concentration was found at the site receiving the accumulated groundwater flow. The lower PFAS concentration was located at the periphery and upgradient points of the field. Several PFAS were detected in crop tissues at the irrigation sites, and the majority of the PFAS accumulated in livestock feed crops were short-chain chemicals [59].

PFAS sorption and transport in soils

PFAS are generally distributed between soil and water after land application of biosolids and irrigation with contaminated water [49,60,61]. Sorption by soils is the determinant to control transport and bioavailability of PFAS to plant root uptake and microbial transformation. In general, PFAS sorption by soils increases with the increase in -CF₂ chain length, and PFAS with sulfonate head group demonstrate a higher sorption than those with carboxylate [62]. The C-F bond is considered hydrophobic but has a high degree of polarity such that the interaction with sorbent surfaces could be considered via polar interactions, which could be stronger than the nonpolar interaction (e.g., van der Waals forces) for other hydrophobic organic compounds [63]. In addition, PFAS with charged head functional groups such as carboxylate or sulfonate are capable of developing electrostatic interaction with anion exchange sites on soil surfaces, particularly for highly weathered soils [64]. Given these special properties, PFAS are expected to exhibit a moderate partitioning to soil organic matter and electrostatic interaction with positively charged sites on soil surfaces (e.g., from Fe/Al oxyhydroxides). Li et al. measured sorption of PFAS by soils and found that protein content, iron oxide content, and anion exchange capacity manifest a greater influence on PFAS sorption than soil organic matter content [65], albeit more studies are needed to further confirm the conclusion.

The influence of specific sorption, for example, cation-bridging interaction or anion exchange, on the overall sorption could be significant, but the quantity remains nearly unknown [64]. Wang et al. prepared several soils via occupying soil-cation-exchange sites with mono-, di-, and tri-valent cations (i.e., K⁺, Ca²⁺, and Fe³⁺) and measured sorption of several PFCAs by the soils. Sorption of PFCA by Ca²⁺ and Fe³⁺ soils was significantly greater than that by K⁺-soils, suggesting the substantial contributions of cation-bridging interaction to the overall sorption [66]. Soil solution pH, ionic strength, and composition also affect PFAS sorption to metal oxides in soils. For example, sorption of PFOA and PFOS by boehmite (AlO(OH)) decreased with pH increase from 4 to 7, due mainly to the reduction in positive surface charge density on the mineral surfaces at higher pH [67,68]. Similar results have been observed for PFAS sorption by the soils containing high content of Fe and Al oxides [69]. Chen et al. reported the increase in PFOA and PFOS sorption with increasing ionic strength in solution because the compressed electric double layer facilitates PFAS interaction with positive charges on the tropical soil surfaces [67]. The presence of divalent cation Mg²⁺ and Ca²⁺ in solution could enhance PFAS sorption by soils [64,70,71]. The attempts to relate PFAS sorption to bulk soil properties such as soil organic matter, clay content, or pH alone fail to explain PFAS sorption behaviors across multiple studies and soil types [64]. There are still large knowledge gaps in understanding fundamental sorption mechanism of PFAS by soils at the molecular scale, which hinders the development of models to predict PFAS transport in soils.

PFAS residues in surface soils could be desorbed and dissolved in the infiltrating water from precipitation, irrigation, runoff and stormwater, leach down to aquifer and transport with the groundwater flow. The movement of PFAS with soil depth is related to their sorption by soils, as well as water flow rate and volume. Long-chain PFAS, strongly sorbed by soils, represent the majority of PFAS remaining within the shallow soil profile, whereas short-chain PFAS occupy the majority of PFAS in soils at deep depth. This distribution pattern has been observed at many PFAS-contaminated sites owing to land application of biosolids and use AFFF formulations at firefighting sites [39,47,49,72,73]. For example, Brusseau et al. examined PFOS concentration at the U.S. Air Force AFFF-impacted sites and showed that PFOS concentration decrease exponentially by several orders of magnitude with soil depth [38].

In addition to the impact of sorption to transport, the strong accumulation of PFAS at air–water interfaces also retards their migration in soil profiles. PFAS are amphiphilic compounds and can reduce the surface tension at the air–water interfaces, leading to the enhanced accumulation. Rain precipitation increases soil water content and simultaneously decreases PFAS

retention owing to the reduction in the air–water interfacial area [74–76]. The magnitude of air–water interfacial adsorption and transport in the unsaturated soils is related to PFAS chemical structures. The contribution of air–water interfacial accumulation to the total sorption increases with PFAS chain length [77]. This result suggests that sorption and air–water interfacial accumulation both contribute to the retention of long-chain PFAS in the top-layer soils. This is consistent with the observations at many AFFF-impacted sites. PFAS transport in soils is influenced not only by their sorption by soils, but also by the changes in soil water content due to drainage, infiltration, and evapotranspiration, which could particularly impact long-chain PFAS over a long-term time frame.

At many sites, such as AFFF application to fuel-spilling fire, PFAS migrate with organic co-contaminants to groundwater, and the co-contaminants could influence PFAS transport. For the instance of AFFF-impacted sites, the intentional use or accidental release of hydrocarbon fuels and chlorinated solvents during firefighting training events and emergency response commonly results in subsurface co-contamination with nonaqueous-phase liquids (NAPLs). PFAS partitioning in the hydrocarbon fuel NAPLs could alter the distribution and transport in subsurface [78,79]. The NAPL could retard transport of long-chain PFAS in soils and subsurface via partitioning in the organic liquid phase and accumulation at the NAPL–water interfaces [80,81]. The later mechanism plays the major role in determining long-chain PFAS transport behaviors in soils. For example, Liao et al. demonstrated that the accumulation of PFOS and PFNA at NAPL–water interface accounted for ~90% of their retardation at the NAPL saturation of ~16%, while short-chain PFAS (i.e., PFBS and PFHpA) were highly mobile even in the presence of residual NAPL [82].

Remediation of PFAS contamination

The wide distribution of PFAS in lands and migration to the surrounding water bodies cause damages to ecosystems and human health [4,34–37]. There is an urgent need to develop effective and environmentally acceptable remediation technologies to mitigate the impacts of PFAS in soils. Currently, remediation technologies predominantly target water contamination with PFAS. The treatment of PFAS-contaminated soils is limited to the mobilization or immobilization strategies. The immobilization is to sequester PFAS in soils using geosorbents with strong sorption capacity; the purpose is to minimize PFAS mobility in soils and reduce their bioavailability to the surrounding biota [83]. The common-use geosorbents are activated carbons [84], organic resin [85], biochars [86–88], modified clays [89], and Fe/Al oxides [90,91]. The applications of individual or mixture of these geosorbents could effectively sequester PFAS in soils because of their strong sorption capacity for

PFAS [92]. The application of this remediation technology is to trap PFAS in soils and minimize their transport in soil. However, the major challenge is that PFAS still remain in soils, and the long-term effects of the immobilized PFAS in soils are nearly unknown.

Several types of geosorbents have been tested for applications in laboratory experiments and contaminated field for sequestration of PFAS in soils. Among these geosorbents, activated carbons, either in powder or granular form, are the most common sorbents or a major component in the soil amendment mixture. In a laboratory study with soils collected from firefighting training sites at several airports in Norway, the activated carbon-amended soils at the rate of 4% demonstrated very efficient sequestration of PFAS in soils; no detectable PFOA, PFOS, or PFHxS was found in the associated water [93]. Activated carbon amendment (3%) could achieve the reduction of PFAS leaching by 94–99.9% from soils [94]. Activated carbons as sorbents could effectively immobilize PFAS in soils, and this method is more efficient in application to the soils of more coarse structures and less organic matter content.

Soil washing, either in-situ or ex-situ, is a mobilization method and offers an effective approach to remove PFAS from soils. The principle of soil washing is to use washing solution (often containing chemical additives) to aid PFAS desorption from soils, mobilization, and solubilization in water. There is very limited information on soil washing for PFAS removal available in the published records as a remediation strategy. Several recent remediation reports showed that soil washing with water recirculation could be an effective approach to remove PFAS from soils including those with high clay content [95–97]. For example, Grimison et al. used soil washing plant settings to treat soils collected from a firefighting training site. The procedure effectively removed PFAS from soils with the average removal rates of up to 97% for PFCA and 95% for PFSA (Grimison et al., 2023). The lower removal efficiency was observed in some in-situ soil washing trials. Hoisater et al. conducted in-situ soil washing demonstration at a firefighting training site, and the removal efficiency was estimated between 23 and 73% (with a wide variation) for PFOS [96].

The generated wastewater from soil washing contains PFAS and many other types of contaminants, which could be either concentrated into a small volume/mass or destroyed using chemical methods. Membrane technology such as reverse osmosis and nanofiltration can efficiently separate PFAS and other contaminants from the wastewater [98–100]. The separated clean water could be reused for the circular soil washing, while the concentrated PFAS waste needs further treatment, and destruction is preferred. Currently, activated carbon and ion exchange resins are the most common sorbents to remove PFAS from water [101–106]. Chemical

oxidation processes have been utilized to destruct PFAS in water [35,107]. The concern about chemical oxidation includes the generation of more mobile short-chain PFAS. In addition, chemical oxidation is not very effective for PFAS compounds because the highly electronegative F-atom could envelop and protect the carbon skeleton from the attack by oxidative radicals ($\cdot\text{OH}$). Chemical reduction has been investigated as a strategy to destroy PFAS because C–F is more amenable to reductive reactions. Some reductive processes have been demonstrated to be effective to destroy PFAS including UV-irradiated sulfite, UV-irradiated-iodide, and UV-irradiated-dithionite in solution [108–114]. However, the efficacy of these processes could be substantially diminished in the water containing dissolved oxygen and nitrate due to consumption of reductive species. There are also concerns regarding the environmental impacts of using iodides or sulfites in water treatment.

Research perspectives

The large amount of manufactured PFAS and their diverse applications in industrial production and household activities, combined with their nature of high stability, mobility and persistence, lead to their wide distribution in the soil environment. This review summarizes the status of knowledge on the sources of PFAS in soils, sorption and transport behaviors, and several remediation strategies. The primary sources for disseminating PFAS in soils originate from utilization of AFFF formulations at firefighting sites and land application of biosolids. PFAS in soils could leach down to groundwater, and the extent of migration is governed by PFAS sorption by soils, water content, and co-contamination. Although some PFAS-contaminated sites have been selected for remediation, not much information was documented, which limits the comparison and evaluation of the performance of specific remediation technologies. To alleviate PFAS impact to soil and human health, environmental fate and remediation of PFAS in soils warrant more thorough investigations with some suggested research priorities worthy of immediate attention.

- (1) For the sources and types of PFAS in soils, future studies should aim to investigate more pathways such as air deposition, rainfall, industrial emission, water recycling, and pesticide application as potential sources of PFAS into soils. Both target and nontarget PFAS analysis should be conducted and include major precursors that may serve as long-term release sources of mobile PFAS.
- (2) For sorption and transport of PFAS in soils, more mechanistic studies are needed to better understand sorption of PFAS at soil–water–air interfaces. Based

on the fundamental understanding, quantitative models for PFAS sorption are expected via correlating with specific soil properties, interfacial parameters, PFAS chemical structures, and solution chemistry. Further evaluation of sorption models should be extended to the low concentration of PFAS in soils and more complicated environmental scenarios, for example, the sites in the presence of co-contaminants under unsaturated condition.

- (3) PFAS immobilization in soils using geosorbent amendments could be an emergency management protocol for large contaminated areas such as agricultural lands. This management procedure has demonstrated as an effective approach to sequester PFAS in soils, reduce the migration of PFAS with water flow, as well as to mitigate their bioavailability to plant uptake. However, it is not clear about the stability of the sequestered PFAS in the amendments. More studies are needed to evaluate the potential release of the sequestered PFAS to water and the long-term impacts to the environmental quality and ecosystem health.
- (4) The better understanding of PFAS sorption in soils helps the selection of optimal solution in the remediation using soil washing or flushing technology. PFAS in the elution could be treated with chemical destruction methods. The combination of mobilization techniques with destruction approach should be worthy of particular efforts since this synergy could offer a cost-effective treatment train for PFAS in soils.

CRedit authorship contribution statement

Hui Li: Conceptualization, Methodology, Investigation, Supervision, Funding acquisition, Writing – original draft, Writing – review & editing. **Pulane Koosaletse-Mswela:** Conceptualization, Investigation, Writing – review & editing.

Declaration of competing interest

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Data availability

No data was used for the research described in the article.

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References

Papers of particular interest, published within the period of review, have been highlighted as:

* of special interest

1. CDC: *Fourth national report on human exposure to environmental chemicals*. Centers for Disease Control and Prevention; 2015.
2. Guelfo JL, Adamson DT: **Evaluation of a national data set for insights into sources, composition, and concentrations of per- and polyfluoroalkyl substances (PFASs) in U.S. drinking water**. *Environ Pollut* 2018, **236**:505–513.
3. Sunderland EM, Hu XDC, Dassuncao C, Tokranov AK, Wagner CC, Allen JG: **A review of the pathways of human exposure to poly- and perfluoroalkyl substances (PFASs) and present understanding of health effects**. *J Expo Sci Environ Epidemiol* 2019, **29**:131–147.
4. Backe WJ, Day TC, Field JA: **Zwitterionic, cationic, and anionic fluorinated chemicals in aqueous film forming foam formulations and groundwater from US military bases by nonaqueous large-volume injection HPLC-MS/MS**. *Environ Sci Technol* 2013, **47**:5226–5234.
5. Brambilla G, D'Hollander W, Oliaei F, Stahl T, Weber R: **Pathways and factors for food safety and food security at PFOS contaminated sites within a problem based learning approach**. *Chemosphere* 2015, **129**:192–202.
6. Filipovic M, Woldegiorgis A, Norstrom K, Bibi M, Lindberg M, Osteras A-H: **Historical usage of aqueous film forming foam: a case study of the widespread distribution of perfluoroalkyl acids from a military airport to groundwater, lakes, soils and fish**. *Chemosphere* 2015, **129**:39–45.
7. Kannan K: **Perfluoroalkyl and polyfluoroalkyl substances: current and future perspectives**. *Environ Chem* 2011, **8**: 333–338.
8. Prevedouros K, Cousins IT, Buck RC, Korzeniowski SH: **Sources, fate and transport of perfluorocarboxylates**. *Environ Sci Technol* 2006, **40**:32–44.
9. Calafat AM, Wong L-Y, Kuklenyik Z, Reidy JA, Needham LL: **Polyfluoroalkyl chemicals in the US population: data from the national health and nutrition examination survey (NHANES) 2003-2004 and comparisons with NHANES 1999-2000**. *Environ Health Perspect* 2007, **115**:1596–1602.
10. Codling G, Hosseini S, Corcoran MB, Bonina S, Lin T, Li A, Sturchio NC, Rockne KJ, Ji K, Peng H, Giesy JP: **Current and historical concentrations of poly and perfluorinated compounds in sediments of the northern Great Lakes - superior, Huron, and Michigan**. *Environ Pollut* 2018, **236**:373–381.
11. Dalahmeh S, Tirgani S, Komakech AJ, Niwagaba CB, Ahrens L: **Per- and polyfluoroalkyl substances (PFASs) in water, soil and plants in wetlands and agricultural areas in Kampala, Uganda**. *Sci Total Environ* 2018, **631–632**:660–667.
12. Giesy JP, Kannan K: **Global distribution of perfluorooctane sulfonate in wildlife**. *Environ Sci Technol* 2001, **35**:1339–1342.
13. Houtz E, Wang M, Park J-S: **Identification and fate of aqueous film forming foam derived per- and polyfluoroalkyl substances in a wastewater treatment plant**. *Environ Sci Technol* 2018, **52**:13212–13221.
14. Kannan K, Corsolini S, Falandysz J, Fillmann G, Kumar KS, Loganathan BG, Mohd MA, Olivero J, Van Wouwe N, Yang JH, Aldous KM: **Perfluorooctanesulfonate and related fluorochemicals in human blood from several countries**. *Environ Sci Technol* 2004, **38**:4489–4495.
15. Stahl T, Riebe RA, Falk S, Failing K, Brunn H: **Long-term lysimeter experiment to investigate the leaching of perfluoroalkyl substances (PFASs) and the carry-over from soil to plants: results of a pilot study**. *J Agric Food Chem* 2013, **61**: 1784–1793.
16. Taniyasu S, Kannan K, Horii Y, Hanari N, Yamashita N: **A survey of perfluorooctane sulfonate and related perfluorinated organic compounds in water, fish, birds, and humans from Japan**. *Environ Sci Technol* 2003, **37**:2634–2639.
17. Biegel LB, Hurtt ME, Frame SR, O'Connor JC, Cook JC: **Mechanisms of extrahepatic tumor induction by peroxisome proliferators in male CD rats**. *Toxicol Sci* 2001, **60**:44–55.
18. Frisbee SJ, Shankar A, Knox SS, Steenland K, Savitz DA, Fletcher T, Ducatman AM: **Perfluorooctanoic acid, perfluorooctanesulfonate, and serum lipids in children and adolescents**. *Arch Pediatr Adolesc Med* 2010, **164**:860–869.
19. Gebbink WA, Hebert CE, Letcher RJ: **Perfluorinated carboxylates and sulfonates and precursor compounds in herring gull eggs from colonies spanning the Laurentian Great Lakes of North America**. *Environ Sci Technol* 2009, **43**:7443–7449.
20. Lau C, Thibodeaux JR, Hanson RG, Narotsky MG, Rogers JM, Lindstrom AB, Strynar MJ: **Effects of perfluorooctanoic acid exposure during pregnancy in the mouse**. *Toxicol Sci* 2006, **90**:510–518.
21. DeWitt JC: *Toxicological effects of perfluoroalkyl and polyfluoroalkyl substances*. New York: Humana Press, Springer International Publishing; 2015.
22. Lorber M, Egeghy PP: **Simple intake and pharmacokinetic modeling to characterize exposure of Americans to perfluorooctanoic acid, PFOA**. *Environ Sci Technol* 2011, **45**: 8006–8014.
23. ATSDR: *Toxicological profile of perfluoroalkyls agency for toxic substances and disease registry*, Atlanta, GA. 2018.
24. Bolan N, Sarkar B, Vithanage M, Singh G, Tsang DCW, Mukhopadhyay R, Ramadass K, Vinu A, Sun YQ, Ramanayaka S, Hoang SA, Yan YB, Li Y, Rinklebe J, Li H, Kirkham MB: **Distribution, behaviour, bioavailability and remediation of poly- and per-fluoroalkyl substances (PFAS) in solid biowastes and biowaste-treated soil**. *Environ Int* 2021, **155**:106600.
25. Bolan N, Sarkar B, Yan YB, Li Q, Wijesekera H, Kannan K, Tsang DCW, Schauerte M, Bosch J, Noll H, Ok YS, Scheckel K, Kumpiene J, Gobindal K, Kah M, Sperry J, Kirkham MB, Wang HL, Tsang YF, Hou DY, Rinklebe J: **Remediation of poly- and perfluoroalkyl substances (PFAS) contaminated soils - to mobilize or to immobilize or to degrade?** *J Hazard Mater* 2021, **401**:123892.
26. Blaine AC, Rich CD, Sedlacko EM, Hundal LS, Kumar K, Lau C, Mills MA, Harris KM, Higgins CP: **Perfluoroalkyl acid distribution in various plant compartments of edible crops grown in biosolids-amended soils**. *Environ Sci Technol* 2014, **48**: 7858–7865.
27. Blaine AC, Rich CD, Sedlacko EM, Hyland KC, Stushnoff C, Dickenson ERV, Higgins CP: **Perfluoroalkyl acid uptake in lettuce (*Lactuca sativa*) and strawberry (*Fragaria ananassa*) irrigated with reclaimed water**. *Environ Sci Technol* 2014, **48**: 14361–14368.
28. Gallen C, Drage D, Kaserzon S, Baduel C, Gallen M, Banks A, Broomhall S, Mueller JF: **Occurrence and distribution of brominated flame retardants and perfluoroalkyl substances in Australian landfill leachate and biosolids**. *J Hazard Mater* 2016, **312**:55–64.
29. Sinclair E, Kannan K: **Mass loading and fate of perfluoroalkyl surfactants in wastewater treatment plants**. *Environ Sci Technol* 2006, **40**:1408–1414.
30. Stahl LL, Snyder BD, Olsen AR, Kincaid TM, Wathen JB, McCarty HB: **Perfluorinated compounds in fish from US urban rivers and the Great Lakes**. *Sci Total Environ* 2014, **499**: 185–195.
31. Strynar MJ, Lindstrom AB, Nakayama SF, Egeghy PP, Helfant LJ: **Pilot scale application of a method for the analysis of perfluorinated compounds in surface soils**. *Chemosphere* 2012, **86**:252–257.
32. Taniyasu S, Yamashita N, Moon HB, Kwok KY, Lam PK, Horii Y, Petrick G, Kannan K: **Does wet precipitation represent local**

- and regional atmospheric transportation by perfluorinated alkyl substances? *Environ Int* 2013, **55**:25–32.
33. Tian Y, Yao Y, Chang S, Zhao Z, Zhao Y, Yuan X, Wu F, Sun H: **Occurrence and phase distribution of neutral and ionizable per- and polyfluoroalkyl substances (PFASs) in the atmosphere and plant leaves around landfills: a case study in tianjin, China.** *Environ Sci Technol* 2018, **52**: 1301–1310.
 34. Anderson RH, Long GC, Porter RC, Anderson JK: **Occurrence of select perfluoroalkyl substances at US Air Force aqueous film-forming foam release sites other than fire-training areas: field-validation of critical fate and transport properties.** *Chemosphere* 2016, **150**:678–685.
 35. Houtz EF, Higgins CP, Field JA, Sedlak DL: **Persistence of perfluoroalkyl acid precursors in AFFF-impacted ground-water and soil.** *Environ Sci Technol* 2013, **47**:8187–8195.
 36. Moody CA, Field JA: **Perfluorinated surfactants and the environmental implications of their use in fire-fighting foams.** *Environ Sci Technol* 2000, **34**:3864–3870.
 37. Nickerson A, Maizel AC, Kulkarni PR, Adamson DT, Kornuc JJ, Higgins CP: **Enhanced extraction of AFFF-associated PFASs from source zone soils.** *Environ Sci Technol* 2020, **54**: 4952–4962.
 - [38. * Brusseau ML, Anderson RH, Guo B: **PFAS concentrations in soils: background levels versus contaminated sites.** *Sci Total Environ* 2020, **740**:140017.
- PFAS are widely distributed in soils globally with concentration up to mg/kg levels, based on the analysis of data collected from >2500 sites throughout of the world.
- [39. * Baduel C, Mueller JF, Rotander A, Corfield J, Gomez-Ramos MJ: **Discovery of novel per- and polyfluoroalkyl substances (PFASs) at a fire fighting training ground and preliminary investigation of their fate and mobility.** *Chemosphere* 2017, **185**:1030–1038.
- Many novel PFAS precursors were identified in the soils at a firefighting site which provided decades of training with aqueous film foams.
40. Chu SG, Letcher RJ: **Side-chain fluorinated polymer surfactants in aquatic sediment and biosolid-augmented agricultural soil from the Great Lakes basin of North America.** *Sci Total Environ* 2017, **607**:262–270.
 41. Wen B, Li LF, Zhang HN, Ma YB, Shan XQ, Zhang SZ: **Field study on the uptake and translocation of perfluoroalkyl acids (PFAAs) by wheat (*Triticum aestivum* L.) grown in biosolids-amended soils.** *Environ Pollut* 2014, **184**:547–554.
 42. Association, Northeast Biosolids and Residuals: *A national biosolids regulation, quality, end use and disposal survey*, Tamworth, NH. 2007.
 43. Alder AC, van der Voet J: **Occurrence and point source characterization of perfluoroalkyl acids in sewage sludge.** *Chemosphere* 2015, **129**:62–73.
 44. Loganathan BG, Sajwan KS, Sinclair E, Kumar KS, Kannan K: **Perfluoroalkyl sulfonates and perfluorocarboxylates in two wastewater treatment facilities in Kentucky and Georgia.** *Water Res* 2007, **41**:4611–4620.
 45. Schultz MM, Higgins CP, Huset CA, Luthy RG, Barofsky DF, Field JA: **Fluorochemical mass flows in a municipal wastewater treatment facility.** *Environ Sci Technol* 2006, **40**: 7350–7357.
 - [46. * Venkatesan AK, Halden RU: **National inventory of perfluoroalkyl substances in archived US biosolids from the 2001 EPA National Sewage Sludge Survey.** *J Hazard Mater* 2013, **252**:413–418.
- This study reports the summary of PFAS in the biosolids samples collected from 94 wastewater treatment plants in 32 states and the District of Columbia
- [47. * Washington JW, Yoo H, Ellington JJ, Jenkins TM, Libelo EL: **Concentrations, distribution, and persistence of perfluoroalkylates in sludge-applied soils near Decatur, Alabama, USA.** *Environ Sci Technol* 2010, **44**:8390–8396.
- This paper provides a comprehensive analysis of concentration, fate and persistence of PFAS in soils receiving biosolids.
- [48. * Johnson GR: **PFAS in soil and groundwater following historical land application of biosolids.** *Water Res* 2022, **211**: 118035.
- PFAS from land-applied sewage sludge could release PFAS into soils and leach down to groundwater.
- [49. * Sepulvado JG, Blaine AC, Hundal LS, Higgins CP: **Occurrence and fate of perfluorochemicals in soil following the land application of municipal biosolids.** *Environ Sci Technol* 2011, **45**:8106–8112.
- Land application of typical municipal biosolids for plant fertility values could also disseminate PFAS in agricultural soils.
50. Eriksson U, Haglund P, Karrman A: **Contribution of precursor compounds to the release of per- and polyfluoroalkyl substances (PFASs) from waste water treatment plants (WWTPs).** *J Environ Sci* 2017, **61**:80–90.
 51. Bizkarguenaga E, Zabaleta I, Mijangos L, Iparraguirre A, Fernandez LA, Prieto A, Zuloaga O: **Uptake of perfluorooctanoic acid, perfluorooctane sulfonate and perfluorooctane sulfonamide by carrot and lettuce from compost amended soil.** *Sci Total Environ* 2016, **571**:444–451.
 52. Lee H, Tevlin AG, Mabury SA, Mabury SA: **Fate of polyfluoroalkyl phosphate diesters and their metabolites in biosolids-applied soil: biodegradation and plant uptake in greenhouse and field experiments.** *Environ Sci Technol* 2014, **48**:340–349.
 53. Lindstrom AB, Strynar MJ, Delinsky AD, Nakayama SF, McMillan L, Libelo EL, Neill M, Thomas L: **Application of WWTP biosolids and resulting perfluorinated compound contamination of surface and well water in Decatur, Alabama, USA.** *Environ Sci Technol* 2011, **45**:8015–8021.
 54. Navarro I, de la Torre A, Sanz P, Porcel MA, Pro J, Carbonell G, Martinez ML: **Uptake of perfluoroalkyl substances and halogenated flame retardants by crop plants grown in biosolids-amended soils.** *Environ Res* 2017, **152**:199–206.
 55. O'Connor J, Bolan NS, Kumar M, Nitai AS, Ahmed MB, Bolan SS, Vithanage M, Rinklebe J, Mukhopadhyay R, Srivastava P, Sarkar B, Bhatnagar A, Wang HL, Siddique KHM, Kirkham MB: **Distribution, transformation and remediation of poly- and per-fluoroalkyl substances (PFAS) in wastewater sources.** *Process Saf Environ Protect* 2022, **164**:91–108.
 56. Pan CG, Liu YS, Ying GG: **Perfluoroalkyl substances (PFASs) in wastewater treatment plants and drinking water treatment plants: removal efficiency and exposure risk.** *Water Res* 2016, **106**:562–570.
 57. Kim H, Ekpe OD, Lee JH, Kim DH, Oh JE: **Field-scale evaluation of the uptake of Perfluoroalkyl substances from soil by rice in paddy fields in South Korea.** *Sci Total Environ* 2019, **671**:714–721.
 58. Zhang M, Wang P, Lu YL, Lu XT, Zhang AQ, Liu ZY, Zhang YQ, Khan K, Sarvajayakesavalu S: **Bioaccumulation and human exposure of perfluoroalkyl acids (PFAAs) in vegetables from the largest vegetable production base of China.** *Environ Int* 2020, **135**:105347.
 - [59. * Mroczko O, Preisendanz HE, Wilson C, Mashtare ML, Elliott HA, Veith TL, Soder KJ, Watson JE: **Spatiotemporal patterns of PFAS in water and crop tissue at a beneficial wastewater reuse site in central Pennsylvania.** *J Environ Qual* 2022, **51**: 1282–1297.
- Irrigation of well groundwater originating from treated wastewater results in the dissemination of PFAS in soils and accumulation in plants.
60. Navarro I, de la Torre A, Sanz P, Porcel MA, Carbonell G, Martinez MD: **Transfer of perfluorooctanesulfonate (PFOS), decabrominated diphenyl ether (BDE-209) and Dechlorane Plus (DP) from biosolid-amended soils to leachate and runoff water.** *Environ Chem* 2018, **15**:195–204.
 61. Xiao F, Simcik MF, Halbach TR, Gulliver JS: **Perfluorooctane sulfonate (PFOS) and perfluorooctanoate (PFOA) in soils and groundwater of a US metropolitan area: migration and implications for human exposure.** *Water Res* 2015, **72**:64–74.
 62. Milinovic J, Lacorte S, Vidal M, Rigol A: **Sorption behaviour of perfluoroalkyl substances in soils.** *Sci Total Environ* 2015, **511**:63–71.

63. Higgins CP, Luthy RG: **Sorption of perfluorinated surfactants on sediments.** *Environ Sci Technol* 2006, **40**:7251–7256.
 - [64. Li Y, Oliver DP, Kookana RS: **A critical analysis of published data to discern the role of soil and sediment properties in determining sorption of per and polyfluoroalkyl substances (PFASs).** *Sci Total Environ* 2018, **628–629**:110–120.
- Multiple factors of soil property and solution chemistry could influence PFAS sorption by soils.
65. Li F, Fang X, Zhou Z, Liao X, Zou J, Yuan B, Sun W: **Adsorption of perfluorinated acids onto soils: kinetics, isotherms, and influences of soil properties.** *Sci Total Environ* 2019, **649**: 504–514.
 - [66. Wang W, Rhodes G, Zhang W, Yu XY, Teppen BJ, Li H: **Implication of cation-bridging interaction contribution to sorption of perfluoroalkyl carboxylic acids by soils.** *Chemosphere* 2022, **290**:133224.
- The paper provides quantitative estimates of PFAS sorption by soils through cation-bridging interactions.
67. Chen Y-C, Lo S-L, Li N-H, Lee Y-C, Kuo J: **Sorption of perfluoroalkyl substances (PFASs) onto wetland soils.** *Desalination Water Treat* 2013, **51**:7469–7475.
 68. Shih K, Wang F: **Adsorption behavior of perfluorochemicals (PFCs) on boehmite: influence of solution chemistry.** *Proceedia Environmental Sciences* 2013, **18**:106–113.
 69. Oliver DP, Li Y, Orr R, Nelson P, Barnes M, McLaughlin MJ, Kookana RS: **The role of surface charge and pH changes in tropical soils on sorption behaviour of per- and polyfluoroalkyl substances (PFASs).** *Sci Total Environ* 2019, **673**: 197–206.
 70. Chen H, Zhang C, Yu YX, Han JB: **Sorption of perfluorooctane sulfonate (PFOS) on marine sediments.** *Mar Pollut Bull* 2012, **64**:902–906.
 71. Pereira HC, Ullberg M, Kleja DB, Gustafsson JP, Ahrens L: **Sorption of perfluoroalkyl substances (PFASs) to an organic soil horizon - effect of cation composition and pH.** *Chemosphere* 2018, **207**:183–191.
 72. Casson R, Chiang SY: **Integrating total oxidizable precursor assay data to evaluate fate and transport of PFASs.** *Remediation* 2018, **28**:71–87.
 73. Dauchy X, Boiteux V, Colin A, Hemard J, Bach C, Rosin C, Munoz JF: **Deep seepage of per- and polyfluoroalkyl substances through the soil of a firefighter training site and subsequent groundwater contamination.** *Chemosphere* 2019, **214**:729–737.
 - [74. Brusseau ML, Yan N, Van Glubt S, Wang YK, Chen W, Lyu Y, Dungan B, Carroll KC, Holguin FO: **Comprehensive retention model for PFAS transport in subsurface systems.** *Water Res* 2019, **148**:41–50.
- Several long-chain PFAS demonstrated the greater transport retardation in unsaturated soils due to the increased air–water interfacial adsorption.
75. Lyu Y, Brusseau ML, Chen W, Yan N, Fu XR, Lin XY: **Adsorption of PFOA at the air-water interface during transport in unsaturated porous media.** *Environ Sci Technol* 2018, **52**: 7745–7753.
 76. Silva JAK, Martin WA, McCray JE: **Air-water interfacial adsorption coefficients for PFAS when present as a multi-component mixture.** *J Contam Hydrol* 2021, **236**: 103731.
 77. Lyu Y, Wang BH, Du XQ, Guo B, Brusseau ML: **Air-water interfacial adsorption of C4-C10 perfluorocarboxylic acids during transport in unsaturated porous media.** *Sci Total Environ* 2022, **831**:154905.
 78. Chen H, Chen S, Quan X, Zhao YZ, Zhao HM: **Sorption of perfluorooctane sulfonate (PFOS) on oil and oil-derived black carbon: influence of solution pH and Ca²⁺.** *Chemosphere* 2009, **77**:1406–1411.
 79. Guelfo JL, Higgins CP: **Subsurface transport potential of perfluoroalkyl acids at aqueous film-forming foam (AFFF)-impacted sites.** *Environ Sci Technol* 2013, **47**:4164–4171.
 80. McKenzie ER, Siegrist RL, McCray JE, Higgins CP: **The influence of a non-aqueous phase liquid (NAPL) and chemical oxidant application on perfluoroalkyl acid (PFAA) fate and transport.** *Water Res* 2016, **92**:199–207.
 81. Van Glubt S, Brusseau ML: **Contribution of nonaqueous-phase liquids to the retention and transport of per and polyfluoroalkyl substances (PFAS) in porous media.** *Environ Sci Technol* 2021, **55**:3706–3715.
 - [82. Liao SC, Arshadi M, Woodcock MJ, Saleeba Z, Pinchbeck D, Liu C, Capiro NL, Abriola LM, Pennell KD: **Influence of residual nonaqueous-phase liquids (NAPLs) on the transport and retention of perfluoroalkyl substances.** *Environ Sci Technol* 2022, **56**:7976–7985.
- The presence of NAPLs could enhance the retention of long-chain PFAS in soils, and slow down their migration with water.
83. Zhang WL, Liang YN: **Changing bioavailability of per- and polyfluoroalkyl substances (PFAS) to plant in biosolids amended soil through stabilization or mobilization.** *Environ Pollut* 2022, **308**:119724.
 84. McGregor R: **In Situ treatment of PFAS-impacted groundwater using colloidal activated Carbon.** *Remediation* 2018, **28**: 33–41.
 85. Uriakhil MA, Sidnell T, Fernandez AD, Lee J, Ross I, Bussemaker M: **Per- and poly-fluoroalkyl substance remediation from soil and sorbents: a review of adsorption behaviour and ultrasonic treatment.** *Chemosphere* 2021, **282**: 131025.
 86. Askeland M, Clarke BO, Cheema SA, Mendez A, Gasco G, Paz-Ferreiro J: **Biochar sorption of PFOS, PFOA, PFHxS and PFHxA in two soils with contrasting texture.** *Chemosphere* 2020, **249**:126072.
 87. Guo W, Lu SY, Shi JH, Zhao X: **Effect of corn straw biochar application to sediments on the adsorption of 17 alpha-ethinyl estradiol and perfluorooctane sulfonate at sediment-water interface.** *Ecotoxicol Environ Saf* 2019, **174**:363–369.
 88. Wang ZY, Alinezhad A, Nason S, Xiao F, Pignatello JJ: **Enhancement of per- and polyfluoroalkyl substances removal from water by pyrogenic carbons: tailoring carbon surface chemistry and pore properties.** *Water Res* 2023, **229**: 119467.
 89. Das P, Arias VA, Kambala V, Mallavarapu M, Naidu R: **Remediation of perfluorooctane sulfonate in contaminated soils by modified clay adsorbent-a risk-based approach.** *Water Air Soil Pollut* 2013, **224**:1714.
 90. Aly YH, McInnis DP, Lombardo SM, Arnold WA, Pennell KD, Hatton J, Simcik MF: **Enhanced adsorption of perfluoroalkyl substances for in situ remediation.** *Environ Sci Water Res Technol* 2019, **5**:1867–1875.
 91. Huo JW, Min XP, Dong QQ, Xu SP, Wang Y: **Comparison of Zn-Al and Mg-Al layered double hydroxides for adsorption of perfluorooctanoic acid.** *Chemosphere* 2022, **287**.
 92. Sorengard M, Kleja DB, Ahrens L: **Stabilization and solidification remediation of soil contaminated with poly- and perfluoroalkyl substances (PFASs).** *J Hazard Mater* 2019, **367**: 639–646.
 93. Kupryianchyk D, Hale SE, Breedveld GD, Cornelissen G: **Treatment of sites contaminated with perfluorinated compounds using biochar amendment.** *Chemosphere* 2016, **142**:35–40.
 - [94. Hale SE, Arp HPH, Slinde GA, Wade EJ, Bjorseth K, Breedveld GD, Straith BF, Moe KG, Jartun M, Hoisaeter A: **Sorbent amendment as a remediation strategy to reduce PFAS mobility and leaching in a contaminated sandy soil from a Norwegian firefighting training facility.** *Chemosphere* 2017, **171**:9–18.
- Activated carbon as a sorbent amendment could effectively secure PFAS in soil and reduce their leaching out from soils.
- [95. Grimison C, Knight ER, Nguyen TMH, Nagle N, Kabiri S, Braunig J, Navarro DA, Kookana RS, Higgins CP, McLaughlin MJ, Mueller JF: **The efficacy of soil washing for the remediation of per- and poly-fluoroalkyl substances (PFASs) in the field.** *J Hazard Mater* 2022, **445**:130441.

Soil washing plant was used to effectively remove PFAS (>90%) from soils. The water stream of PFAS-contaminated water was treated with granular activated carbon and ion-exchange resin, and circulated for the soil washing.

- [96. * Hoisaeter A, Arp HPH, Slinde G, Knutsen H, Hale SE, Breedveld GD, Hansen MC: **Excavated vs novel in situ soil washing as a remediation strategy for sandy soils impacted with per- and polyfluoroalkyl substances from aqueous film forming foams.** *Sci Total Environ* 2021, **794**:148763.
First real field scale of in-situ soil washing was conducted for effectively cleaning up PFAS from contaminated sandy soil.
97. Quinlan J, Morrell C, Nagle N, Maynard KG: **Ex situ soil washing to remove PFAS adsorbed to soils from source zones.** *Remediation* 2022, **32**:151–166.
98. Franke V, McCleaf P, Lindegren K, Ahrens L: **Efficient removal of per- and polyfluoroalkyl substances (PFASs) in drinking water treatment: nanofiltration combined with active carbon or anion exchange.** *Environ Sci Water Res Technol* 2019, **5**: 1836–1843.
99. Patterson C, Burkhardt J, Schupp D, Krishnan ER, Dymont S, Merritt S, Zintek L, Kleinmaier D: **Effectiveness of point-of-use/point-of-entry systems to remove per- and polyfluoroalkyl substances from drinking water.** *AWWA Water Sci* 2019, **1**:1–12.
100. Soriano A, Gorri D, Biegler LT, Uriaga A: **An optimization model for the treatment of perfluorocarboxylic acids considering membrane preconcentration and BDD electrooxidation.** *Water Res* 2019, **164**:114954.
101. Deng SB, Yu QA, Huang J, Yu G: **Removal of perfluorooctane sulfonate from wastewater by anion exchange resins: effects of resin properties and solution chemistry.** *Water Res* 2010, **44**:5188–5195.
102. Du Z, Deng S, Bei Y, Huang Q, Wang B, Huang J, Yu G: **Adsorption behavior and mechanism of perfluorinated compounds on various adsorbents-A review.** *J Hazard Mater* 2014, **274**:443–454.
103. Merino N, Qu Y, Deeb RA, Hawley EL, Hoffmann MR, Mahendra S: **Degradation and removal methods for perfluoroalkyl and polyfluoroalkyl substances in water.** *Environ Eng Sci* 2016, **33**:615–649.
104. Zaggia A, Conte L, Falletti L, Fant M, Chiorboli A: **Use of strong anion exchange resins for the removal of perfluoroalkylated substances from contaminated drinking water in batch and continuous pilot plants.** *Water Res* 2016, **91**:137–146.
105. Zhi Y, Liu J: **Adsorption of perfluoroalkyl acids by carbonaceous adsorbents: effect of carbon surface chemistry.** *Environ Pollut* 2015, **202**:168–176.
106. Zhi Y, Liu J: **Surface modification of activated carbon for enhanced adsorption of perfluoroalkyl acids from aqueous solutions.** *Chemosphere* 2016, **144**:1224–1232.
107. Bruton TK, Sedlak DL: **Treatment of aqueous film-forming foam by heat-activated persulfate under conditions representative of in situ chemical oxidation.** *Environ Sci Technol* 2017, **51**:13878–13885.
108. Chen Z, Teng Y, Wang W, Hong R, Huang L, Wang X, Zhu F, Li H, Hao S, Wu B, Gu C: **Enhanced UV photoreductive destruction of perfluorooctanoic acid in the presence of alcohols: synergistic mechanism of hydroxyl radical quenching and solvent effect.** *Appl Catal B Environ* 2022, **316**:121652.
109. Huang L, Dong W, Hou H: **Investigation of the reactivity of hydrated electron toward perfluorinated carboxylates by laser flash photolysis.** *Chem Phys Lett* 2007, **436**:124–128.
110. Li X, Fang J, Liu G, Zhang S, Pan B, Ma J: **Kinetics and efficiency of the hydrated electron-induced dehalogenation by the sulfite/UV process.** *Water Res* 2014, **62**:220–228.
111. Ochoa-Herrera V, Sierra-Alvarez R, Somogyi A, Jacobsen NE, Wysocki VH, Field JA: **Reductive defluorination of perfluorooctane sulfonate.** *Environ Sci Technol* 2008, **42**:3260–3264.
112. Park H, Vecitis CD, Cheng J, Choi W, Mader BT, Hoffmann MR: **Reductive defluorination of aqueous perfluorinated alkyl surfactants: effects of ionic headgroup and chain length.** *J Phys Chem A* 2009, **113**:690–696.
113. Song Z, Tang H, Wang N, Zhu L: **Reductive defluorination of perfluorooctanoic acid by hydrated electrons in a sulfite-mediated UV photochemical system.** *J Hazard Mater* 2013, **262**:332–338.
114. Tian H, Gao J, Li H, Boyd SA, Gu C: **Complete defluorination of perfluorinated compounds by hydrated electrons generated from 3-Indole-acetic-acid in organomodified montmorillonite.** *Sci Rep* 2016, **6**:32949.